

Parameter model — a'_{hd} as a state

$$a_h[k+1] = a_h(h[k+1]) = a_h(h_d[k+1] - \delta h[k+1])$$

$$h_d[k+1] = h_d[k] + \Delta h_d[k]$$

$$\delta h[k+1] = \lambda_c \delta h[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k]$$

True:

$$\begin{aligned} a_h[k+1] &= a_h(h_d[k] + \Delta h_d[k] - \lambda_c \delta h[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k]) \\ &\approx a_h[k] + a'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) a'_{hd}[k] \delta h[k] + a'_{hd}[k] F_{dh}[k] e_{a_h}[k] + a'_{hd}[k] \varepsilon_h[k] \end{aligned}$$

Esti:

$$\begin{aligned} \hat{a}_h[k+1] &= \hat{a}_h[k] + \hat{a}'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] \hat{\delta} h[k] \\ \hat{a}'_{hd}[k+1] &= \hat{a}'_{hd}[k] \end{aligned}$$

Error:

$$\begin{aligned} e_{a_h}[k+1] &= (1 + \hat{a}'_{hd}[k] F_{dh}[k]) e_{a_h}[k] + \Delta h_d[k] e_{a'_h}[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] e_h[k] + \hat{a}'_{hd}[k] \varepsilon_h[k] \\ e_{a'_h}[k+1] &= e_{a'_h}[k] \quad \text{or being updated according to the ratio between two variances (p. 2)} \end{aligned}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{a}'_{hd}[k] & 1 + \hat{a}'_{hd}[k] F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\underline{Q_{55}[k]}$$

$$a'_{hd}[k+1] = a'_{hd}(h[k+1]) \approx a'_{hd}[k] + \underbrace{a''_h[k] \Delta h[k]}_{w_{a'}[k]}, \quad \Delta h[k] = h[k+1] - h[k]$$

$$a''_h \text{ unknown} \Rightarrow w_{a'} \rightarrow Q: \quad Q_{55}[k] = \left(\max |w_{a'}[k]| \right)^2$$

Bound of a''_h ($a_h(h_{wall}) = 0$, $a_h(h_d) = \hat{a}_h$, smooth monotone; $d := h_d - h_{wall}$, h_{wall} known):

$$a_h \propto d^p \Rightarrow a'_h = p \frac{a_h}{d}, \quad a''_h = p(p-1) \frac{a_h}{d^2}$$

$$p = \frac{1}{2}: a''_h = -\frac{1}{4} \frac{a_h}{d^2}; \quad p = 1: a''_h = 0; \quad p = 2: a''_h = 2 \frac{a_h}{d^2}$$

$$\boxed{|a''_h| \leq \kappa \frac{\hat{a}_h}{d^2}, \quad \kappa := \frac{|a''_h| d^2}{a_h}}$$

$$\kappa \in [0.10, 0.48] \quad (\text{true } c_{\perp}, \bar{h} \in [1.05, 22], \text{ max at } \bar{h} \approx 2.1) \Rightarrow \kappa := 1$$

$$Q_{55} (|\Delta h| \rightarrow |\Delta h_d|):$$

$$\max |w_{a'}[k]| = \kappa \frac{\hat{a}_h[k]}{d^2[k]} |\Delta h_d[k]|$$

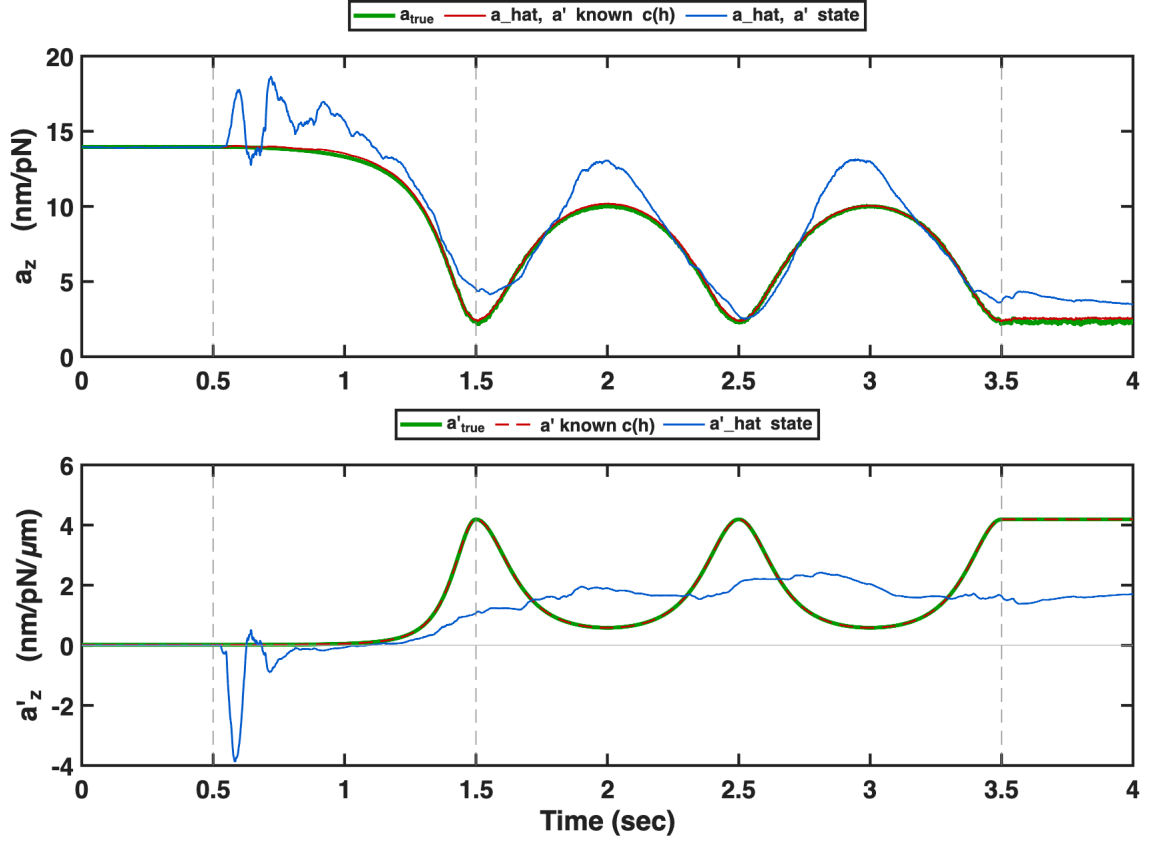
$$\boxed{Q_{55}[k] = \kappa^2 \frac{\hat{a}_h^2[k]}{d^4[k]} \Delta h_d^2[k], \quad \Delta h_d = 0 \Rightarrow Q_{55} = 0}$$

Full process noise ($q_3 = -\varepsilon_h$, $q_4 = \hat{a}'_{hd} \varepsilon_h$: one sample $\Rightarrow$ rank-1; $q_5 = w_{a'}$ independent):

$$\boxed{Q(3:5, 3:5) = \begin{bmatrix} \sigma_{\varepsilon_h}^2 & -\hat{a}'_{hd} \sigma_{\varepsilon_h}^2 & 0 \\ -\hat{a}'_{hd} \sigma_{\varepsilon_h}^2 & \hat{a}_{hd}^2 \sigma_{\varepsilon_h}^2 & 0 \\ 0 & 0 & Q_{55}[k] \end{bmatrix}, \quad \sigma_{\varepsilon_h}^2 = (1 + 2(1 - \lambda_c)^2) \cdot 4k_B T \hat{a}_h}$$

Impl. (current settings): $\kappa = 1$ (`aprime_state_kappa`); $P_{55}[0] = (0.1 a_{nom}/R)^2$ (`aprime_state_P0`; design rule $(\kappa \hat{a}_h[0]/d[0])^2$, same order); learning from $t = 0$ (`aprime_learn_t0` = 0; the freeze variant is rejected — p.3).

Verification — a'_{hd} as a state (z axis, 1 Hz osc)



$\hat{a}_h$ (top) and a'_{hd} (bottom): a' known $c(\bar{h})$ vs a' -as-state (both taylor, learning from $t = 0$; seed 1, study: 20 seeds). Study (osc window): known a' $-0.69\% / 3.08\%$ (bias/std); a' -as-state $+6.31\% / 20.7\%$, slope mean -8.6% , trough peaks $\sim 1/3$ captured, $P_{55}/\text{Var} = 0.05$. Freeze variant (learn released at 1.5 s): $+44.9\% / 74.3\%$, det run diverges — frozen a'_{hd} also freezes the gain feedforward.